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FIRST FIRE OPERATION FOR OVONIC THRESHOLD SWITCH SELECTOR

Abstract

First fire operations for an ovonic threshold switch (OTS) selector is provided. A first fire operation includes setting a peak amplitude of a voltage pulse, and performing at least one cycle, including: providing the voltage pulse to the OTS selector; sensing an output current passing through the OTS selector in response to the received voltage pulse; comparing a peak amplitude of the voltage pulse with a maximum peak amplitude ensuring initialization of the OTS selector; ending the first fire operation if the peak amplitude reaches the maximum peak amplitude; comparing the output current with a target current indicative of initialization of the OTS selector if the peak amplitude is lower than the maximum peak amplitude; ending the first fire operation if the output current reaches the target current; and setting another voltage pulse with a greater peak amplitude if the output current is lower than the target current.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of and claims the priority benefit of a prior application Ser. No. 18/581,340, filed on Feb. 19, 2024, and the prior application Ser. No. 18/581,340 is a continuation application and claims the priority benefit of another prior application Ser. No. 17/826,180, filed on May 27, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

[0002] With advances in digital technology, there is a greater demand for a nonvolatile memory device with higher capacity, less writing power, higher writing/reading speed, and longer service life. In order to meet the demand, refinement of flash memory has been progressed. On the other hand, nonvolatile memory device including memory cells using resistance variable elements for storing digital data has been researched and developed.

[0003] Mostly, each of these nonvolatile memories has field effect transistors (FETs) that control connection between the resistance variable elements and a driving circuit. The FETs have high on/off ratio and prevent leakage current from passing through the unselected memory cells. However, since a FET is a three-terminal device, controlling access of the resistance variable elements by the FETs can significantly limit design flexibility and integration level in creating these nonvolatile memories.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1A is a schematic three-dimensional view illustrating a memory array, according to some embodiments of the present disclosure.

[0006] FIG. 1B is a cross-sectional view schematically illustrating a memory cell in the memory array as shown in FIG. 1A, according to some embodiments of the present disclosure.

[0007] FIG. 1C is an equivalent circuit of the memory cell as shown in FIG. 1B, according to some embodiments of the present disclosure.

[0008] FIG. 1D is a current-voltage (I-V) curve illustrating an operation cycle and an initialization operation of a selector in the memory cell, according to some embodiments of the present disclosure.

[0009] FIG. 2A is a schematic diagram illustrating a voltage pulse used in a first fire operation, according to some embodiments of the present disclosure.

[0010] FIG. 2B is a schematic diagram illustrating the first fire operation performed on a test structure.

[0011] FIG. 2C is a diagram showing dependency of first fire voltage with respect to process temperature of the first fire operation shown in FIG. 2B.

[0012] FIG. 2D is a diagram illustrating dependency of off current of the selector with respect to the width of the voltage pulse provided to the selector.

[0013] FIG. 3A is a flow diagram illustrating an adaptive first fire operation performed on a memory cell, according to some embodiments of the present disclosure.

[0014] FIG. 3B illustrates input voltage pulses and resulted output current during the adaptive first fire operation as shown in FIG. 3A.

[0015] FIG. 4A is a flow diagram illustrating an adaptive first fire operation performed on a memory cell, according to some embodiments of the present disclosure.

[0016] FIG. 4B illustrates input voltage pulses and resulted output current during the adaptive first fire operation as shown in FIG. 4A.

[0017] FIG. 5A is a schematic diagram illustrating an adaptive first fire operation, according to some embodiments of the present disclosure.

[0018] FIG. 5B is a flow diagram illustrating the adaptive first fire operation, according to some embodiments of the present disclosure.

[0019] FIG. 5C illustrates input voltage pulses and resulted output current during the adaptive first fire operation **500** as shown in FIG. 5B.

[0020] FIG. 6 is a schematic cross-sectional view illustrating a semiconductor chip with the memory array (shown in FIG. 1A) embedded therein, according to some embodiments of the present disclosure.

[0021] FIG. 7 is a schematic three-dimensional view illustrating a memory array, according to some embodiments of the present disclosure.

[0022] FIG. 8A is a cross-sectional view schematically illustrating a memory cell, according to some embodiments of the present disclosure.

[0023] FIG. 8B is a cross-sectional view schematically illustrating a memory cell, according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0024] The following disclosure provides many different embodiments or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0025] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0026] FIG. 1A is a schematic three-dimensional view illustrating a memory array **10**, according to some embodiments of the present disclosure.

[0027] Referring to FIG. 1A, the memory array **10** includes memory cells **100** arranged along columns and rows. The memory cells **100** in each column are arranged along a direction Y, while the memory cells **100** in each row are arranged along a direction X intersected with the direction Y. The memory cells **100** are defined at intersections of first signal lines SL1 and second signal lines SL2 running over and intersecting with the first signal lines SL1. The first signal lines SL1 may be connected to or functioned as bottom terminals of the memory cells **100**, while the second signal lines SL2 may be connected to or functioned as top terminals of the memory cells **100**. The first signal lines SL1 may be referred as bit lines, while the second signal lines SL2 may be referred as word lines. Alternatively, the first signal lines SL1 may be referred as word lines, while the second signal lines SL2 may be referred as bit lines. In some embodiments, as shown in FIG. 1A, the first signal lines SL1 extend along the direction Y, while the second signal lines SL2 extend along the direction X. In these embodiments, each first signal line SL1 may connect the bottom terminals of a column of the memory cells **100**, or functioned as a common bottom terminal for a column of the memory cells **100**. In addition, each second signal line SL2 may connect the top terminals of a row of the memory cells **100**, or functioned as a common top terminal for a row of the memory cells **100**. However, the extending directions of the first and second signal lines SL1, SL2 may be swapped or altered, as long as the memory cells **100** are formed at intersections of the first and second signal lines SL1, SL2.

[0028] FIG. 1B is a cross-sectional view schematically illustrating a memory cell **100** in the memory array **10**, according to some embodiments of the present disclosure.

[0029] Referring to FIG. 1A and FIG. 1B, each memory cell **100** may include a selector **110** and a resistance variable storage element **120**. The selector **110** is a two-terminal switching device, and one of the terminals of the selector **110** is shared with the resistance variable storage element **120**. In some embodiments, an electrode **112** is functioned as a top terminal of the selector **110**, and may be coupled to the second signal line SL2 through a conductive via CV. In addition, an electrode **114** may be functioned as a bottom terminal of the selector **110**, and may be shared with the resistance variable storage element **120**. The electrodes **112**, **114** are each formed of a conductive material. As examples, candidates of the conductive material may include Cu, W, TiN, TaN, Ru, AlN, Co, C, the like and combinations thereof.

[0030] A switching layer **116** may be sandwiched between the electrodes **112**, **114**. An electrical resistance across the switching layer **116** may be altered during operation of the selector **110**. When the switching layer **116** is in a low resistance state, the selector **110** is in an on state, and a conduction path can be established through the selector **110**. On the other hand, when the switching layer **116** is in a high resistance state, the selector **110** is in an off state. As the selector **110** is off, only a very limited current could pass through the selector **110**. In some embodiments, the selector **110** is an ovonic threshold switch (OTS) selector. In these embodiments, the selector **110** can be turned on when a voltage bias applied across the switching layer **116** reaches a threshold voltage. On the other hand, when the voltage bias does not reach the threshold voltage or falls below a holding voltage, the conductive path may be cut off, which result in a high resistance state of the switching layer **116** (i.e., an off state of the selector **110**). According to some embodiments, the switching layer **116** is formed of a chalcogenide compound. Candidates of the chalcogenide compound may contain combinations of elements including Te, Se, As, S, Sb or the like. For instance, the switching layer **116** may include GeTe, GeCTe, AsGeSe, GeSbTe, GeSiAsTe, GeSe, GeSbSe, GeSiAsSe, GeS, GeSbS, GeSiAsS, the like, or combinations thereof.

[0031] The resistance variable storage element **120** may be a two-terminal device as well. As described above, the electrode **114** may be functioned as a common terminal of the selector **110** and the resistance variable storage element **120**. In some embodiments, an electrode **122** is functioned as the other terminal of the resistance variable storage element **120**, and may be coupled to the first

signal line SL1. As similar to the electrodes **112**, **114**, the electrode **122** is formed of a conductive material as well. As examples, candidates of the conductive material may include Cu, W, TiN, TaN, Ru, AlN, Co, C, the like and combinations thereof.

[0032] A storage layer **124** lies between the two terminals of the resistance variable storage element **120** (e.g., the electrodes **114**, **122**). Microstructure in the storage layer **124** may be altered according to input signals applied across the storage layer **124**. In corresponding to the microstructure change, the storage layer **124** may be switched between a high resistance state and a low resistance state. Further, the microstructure and thus the resistance state of the storage layer **124** may be held even when the input signal is removed. Accordingly, the resistance variable storage element **120** is a non-volatile memory device. In some embodiments, the memory array **10** including a plurality of the memory cells **100** is a phase change random access memory (PCRAM). In these embodiments, a crystallinity of the storage layer **124** in the resistance variable storage element **120** of each memory cell **100** may be increased when the storage layer **124** is switched to the low resistance state. On the other hand, when the storage layer **124** is in the high resistance state, the storage layer **124** may be amorphous or may have a rather low crystallinity. In these embodiments where the memory array **10** is a PCRAM, the storage layer **124** in each memory cell **100** of the memory array **10** may be formed of a chalcogenide compound. The chalcogenide compound may include Ge, Te and Sb. For instance, the chalcogenide material may be GeSbTe, such as Ge.sub.2Sb.sub.2Te.sub.5 (GST225), Ge.sub.4Sb.sub.2Te.sub.4 (GST424), Ge.sub.4Sb.sub.6Te.sub.7 (GST467) or so forth. As other examples, the chalcogenide may include Ti.sub.0.4Sb.sub.2Te.sub.3, super lattice Sb.sub.2Te.sub.3/TiTe.sub.2, super lattice GeTe/Sb.sub.2Te.sub.3, super lattice Ti.sub.2Te/Sb.sub.2Te.sub.3 or so forth.

[0033] In alternative embodiments, the memory array **10** is a resistive random access memory (RRAM). In these alternative embodiments, when the storage layer **124** in each memory cell **100** of the memory array is at the low resistance, a conductive filament may be formed through the storage layer **124**. On the other hand, such conductive filament may be cut off when the storage layer **124** is switched to the high resistance state. In these embodiments where the memory array **10** is a RRAM, the storage layer **124** in each memory cell **100** of the memory array **10** may be a dielectric layer, such as a high-k dielectric layer. For instance, the dielectric layer may include nickel oxide, titanium oxide, hafnium oxide, zirconium oxide, zinc oxide, tungsten oxide, aluminum oxide, tantalum oxide, molybdenum oxide, copper oxide or so forth.

[0034] In other embodiments, the memory array **10** is a magnetoresistive random access memory (MRAM). In these embodiments, the storage layer **124** in each memory cell **100** of the memory array **10** may be a multilayer structure including an insulating thin layer and magnetic layers lying at opposite sides of the insulating thin layer. The storage layer **124** has a low resistance when alignment of magnetic moments in the magnetic layers is in a parallel configuration. On the other hand, the storage layer **124** has a high resistance when alignment of magnetic moments in the magnetic layers is in an anti-parallel configuration.

[0035] Detailed structure of the resistance variable storage element **120** may vary, according to type, design and manufacturing process of the resistance variable storage element **120**, the present disclosure is not limited thereto.

[0036] FIG. **1C** is an equivalent circuit of a memory cell **100**, according to some embodiments of the present disclosure.

[0037] Referring to FIG. **1B** and FIG. **1C**, the selector **110** is schematically indicated by a switch in the equivalent circuit of the memory cell **100**. Actually, the selector **110** may be bi-directional, and may be alternatively presented by two diodes connected back to back. In addition, the resistance variable storage element **120** is indicated by a variable resistor in the equivalent circuit, since a variable resistor also has an alterable resistance. As shown in FIG. **1C**, the selector **110** and the resistance variable storage element **120** are serially connected between a first signal line SL1 and a second signal line SL2, and share a common terminal. Since the selector **110** and the resistance

variable storage element **120** are connected in series, a voltage applied across the memory cell **100** is divided across the selector **110** and the resistance variable storage element **120**.

[0038] During operation of the memory cell **100**, a voltage pulse may be provided to one of the corresponding first and second signal lines **SL1**, **SL2**, and the other one of these first and second signal lines **SL1**, **SL2** may receive a reference voltage (e.g., a ground voltage). In order to program the resistance variable storage element **120**, the voltage pulse must be high enough for ensuring that a voltage across the selector **110** is greater than the threshold voltage of the selector **110**, such that the selector **110** can be turned on. Accordingly, the resistance variable storage element **120** is accessible, and can be programmed. On the other hand, in some embodiments, the voltage pulse may be provided with a lower peak voltage during a read operation, such that the selector **110** can be turned on when a low resistance state is stored in the resistance variable storage element **120**, and may be in an off state when a high resistance state is stored in the resistance variable storage element **120**. In these embodiments, by detecting whether a conductive path is established through the selector **110** and the resistance variable storage element **120**, the resistance state stored in the resistance variable storage element **120** can be identified. In alternative embodiments, the selector **110** is turned on even during a read operation for sensing a high resistance state stored in the resistance variable storage element **120**.

[0039] FIG. **1D** is a current-voltage (I-V) curve illustrating an operation cycle and an initialization operation of a selector **110**, according to some embodiments of the present disclosure.

[0040] Referring to FIG. **1D**, each operation cycle of the selector **110** may have multiple stages. Along a timeline, a voltage applied across the selector **110** may increase during a first stage **S1**, and then decrease at a third stage **S3**. At the first stage **S1**, a current passing through the selector **110** stays low, and the selector **110** is in a high resistance state (i.e., an off state). When the voltage is raised over a threshold voltage $V_{sub.TH}$ of the selector **110**, operation of the selector **110** enters a second stage **S2**. At the second stage **S2**, current passing through the selector **110** significantly increases as the voltage across the selector **110** is slightly raised or kept at the threshold voltage $V_{sub.TH}$, and the selector **110** is being switched to a low resistance state (i.e., an on state). When the current is saturate, operation of the selector **110** enters the third stage **S3**, at which the current passing through the selector **110** remains high, and the selector **110** is in the low resistance state (i.e., the on state). A fourth stage **S4** is entered as the voltage across the selector **110** is lowered below a holding voltage $V_{sub.H}$, and current passing through the selector **110** decreases dramatically as the voltage applied across the selector **110** is slightly lowered. At the fourth stage **S4**, the selector **110** is being switched back to the high resistance state (i.e., the off state), and an operation cycle may be completed.

[0041] When the selector **110** is turned on at the third stage **S3**, the resistance variable storage element **120** coupled to the selector **110** can be programmed, or a read current may pass through the resistance variable storage element **120** and the selector **110**. On the other hand, the selector **110** returns to the first stage **S1** when a programming operation of the resistance variable storage element **120** or detection of the read current is over. In addition, the selector **110** may stay at the first stage **S1** while not being selected, or in a condition that the resistance variable storage element **120** at a high resistance state is subjected to a read operation.

[0042] Prior to the operation cycles respectively described above, a conductive path may be initially formed across the switching layer **116** of the selector **110** by a first fire operation, which may be performed during chip test. During the first fire operation, a voltage applied to the switching layer **116** is pulled up to a first fire voltage $V_{sub.FF}$. Meanwhile, a current passing through the selector **110** abruptly increases, and the initial conductive path is formed through the selector **110**. After the initially formed conductive path is cut off by pulling down the voltage applied across the selector **110** to the holding voltage $V_{sub.H}$ from the first fire voltage, the operation cycles respectively described above can be performed. During each of these operation cycles, the conductive path may be reconstructed and then cut off.

[0043] The first fire voltage $V_{\text{sub.FF}}$ may be greater than the threshold voltage $V_{\text{sub.TH}}$, which is further higher than the holding voltage $V_{\text{sub.H}}$. In some embodiments, the memory array **10** is integrated with logic circuits in a single device die, and the integration of the memory array **10** and the logic circuits could be easier if the first fire voltage $V_{\text{sub.FF}}$ used for initializing the selectors **110** in the memory array **10** is reduced to be equal to or lower than a power supply voltage (i.e., $V_{\text{sub.DD}}$) of the logic circuits. In addition, the switching layer **116** of the selector **110** may be subjected to significant electrical stress during the first fire operation. Such electrical stress may result in undesired increase of off current defined as a current passing through the selector **110** when a voltage bias applied across the selector **110** is fixed at a level lower than the threshold voltage $V_{\text{sub.TH}}$. Further, such electrical stress may result in great variations of the threshold voltage $V_{\text{sub.TH}}$ and the off current, including cycle-to-cycle and device-device variations.

[0044] Therefore, by lowering the first fire voltage $V_{\text{sub.FF}}$, the integration of the memory array **10** and the logic circuits can be more easily realized. On top of that, the electrical stress resulted from the first fire operation can be reduced, such that the off current of the selector **110** can be lowered, and variations of the threshold voltage $V_{\text{sub.TH}}$ and the off current can be effectively limited. Optimized first fire operation is provided in various embodiments of the present disclosure.

[0045] FIG. 2A is a schematic diagram illustrating a voltage pulse **P1** used in a first fire operation, according to some embodiments of the present disclosure.

[0046] Referring to FIG. 2A, in some embodiments, the first fire operation is performed by providing the voltage pulse **P1** to the selector **110** at an elevated temperature. According to a possible mechanism, the initial conductive path of the selector **110** can be formed when provided energy get over an activation barrier of the initial conductive path. By performing the first fire operation at an elevated temperature, additional energy is provided. In other words, the energy required for overcoming the activation barrier is not solely provided by the voltage pulse **P1**. As a result, the first fire voltage $V_{\text{sub.FF}}$ indicated by a peak amplitude of the voltage pulse **P1** can be lowered. In addition, the voltage pulse **P1** can be provided with a shorter width. According to some embodiments, a substrate (e.g., a wafer) formed with the memory array **10** is heated to a temperature ranging from about 45° C. to about 150° C. during the first fire operation.

[0047] According to some embodiments, the voltage pulse **P1** is a triangular pulse. An amplitude of the voltage pulse **P1** raises sufficiently during a first half of the voltage pulse **P1** for ensuring formation of the initial conductive path in the selector **110**, then drops down in a second half of the voltage pulse **P1**. In order to ensure formation of the initial conductive path, a peak amplitude **VP1** of the voltage pulse **P1** should be great enough for overcoming the activation barrier of the initial conductive path, and is indicative of the first fire voltage $V_{\text{sub.FF}}$. Since additional thermal energy is provided by heating a substrate (e.g., a wafer) formed with the memory array **10** during the first fire operation, the peak amplitude **VP1** of the voltage pulse **P1** used in the first fire operation (which is indicative of the first fire voltage $V_{\text{sub.FF}}$) can be lower than a peak amplitude of a voltage pulse used in a first fire operation performed at room temperature. By lowering the peak amplitude **VP1** of the voltage pulse **P1** (i.e., lowering the first fire voltage $V_{\text{sub.FF}}$), integration of the memory array **10** and logic circuits with low power supply voltage can be much easier. In some embodiments, the memory array **10** is integrated with logic circuits in a single device die, and a power supply voltage of the logic circuits may range from about 1.8V to about 3V. In these embodiments, the peak amplitude **VP1** of the voltage pulse **P1** can be lowered to about 1.7V, while the first fire operation using the voltage pulse **P1** is performed at about 125° C. According to various embodiments, the first fire operation using the voltage pulse **P1** is performed at an elevated temperature ranging from about 45° C. to about 125° C., and the peak amplitude **VP1** of the voltage pulse **P1** may be limited to a range from about 1.8V to about 3V.

[0048] Further, in some embodiments, the voltage pulse **P1** is provided as a transient pulse, which may also be referred to as an alternating current (AC) pulse. As compared to a quasi-static operation using a slowly varying signal (also referred to as a direct current (DC) operation), the

voltage pulse P1 provided as a transient pulse has a much shorter width W1, such that the disclosed first fire operation is rendered more compatible with actual operating conditions of the memory cell **100**. Moreover, as having a shorter pulse width as compared to a signal used in DC operation, the voltage pulse P1 as an AC pulse may result in much less electrical stress. Accordingly, the off current of the selector **110** can be lowered, and variations of the off current as well as the threshold voltage V.sub.TH of the selector **110** can be effectively limited. As an example, the width W1 of the voltage pulse P1 may range from about 40 nanoseconds (ns) to 20 microseconds (μ s), while a width of a long pulse used in DC operation may range from about 100 milliseconds (ms) to about 10 seconds, or greater.

[0049] FIG. 2B is a schematic diagram illustrating a first fire operation performed on a test structure **200**. FIG. 2C is a diagram showing dependency of first fire voltage V.sub.FF with respect to process temperature of the first fire operation shown in FIG. 2B.

[0050] Referring to FIG. 2B, the test structure **200** is similar to a memory cell **100**, except that the selector **110** in the test structure **200** is in serial connection with a load resistor R.sub.load, rather than the resistance variable storage element **120**. A plurality of test structures **200** are provided, and each test structure **200** is subjected to a first fire operation similar to the first fire operation according to various embodiments of the present disclosure. That is, the first fire operation performed on each test structure **200** is executed at an elevated temperature, and includes applying the voltage pulse P1.

[0051] The voltage pulse P1 is provided to a terminal of the selector **110**. In addition, a terminal of the load resistor R.sub.load is coupled to the selector **110**, whereas the other terminal of the load resistor R.sub.load may be coupled to a voltage source (e.g., a ground voltage). By reading a current IR passing through the load resistor R.sub.load, switching of the selector **110** can be observed. In order to verify temperature dependency of the first fire voltage V.sub.FF with respect to temperature, the first fire operations are performed on the test structures **200** at various temperatures. For each temperature, a group of the test structures **200** are subjected to the first fire operation.

[0052] As shown in FIG. 2C, the first fire operations are performed on the test structures **200** at 25° C., 45° C., 65° C., 85° C., 105° C. and 125° C., and the first fire voltages V.sub.FF obtained by analyzing the currents IR passing through the load resistors R.sub.load of the test structures **200** are shown as data dots in the diagram. For each temperature, over 10 of the test structures **200** are subjected to the first fire operation. A median value of the first fire voltages V.sub.FF corresponding to each temperature is shown as a hollow square. According to the result, a negative dependency of the first fire voltage V.sub.FF with respect to the temperature is verified. That is, the first fire voltage V.sub.FF decreases as the temperature of the first fire operation increases. In certain cases, as the temperature of the first fire operation increases from 25° C. to 125° C., the first fire voltage V.sub.FF (median value) may decrease from about 2.2 V to about 1.8 V or lower.

[0053] Moreover, in order to observe dependency of the off current with respect to the width W1 of the voltage pulse V.sub.FF, a first fire operation using the voltage pulse V.sub.FF with a first width W1 of 20 μ s is performed on a test structures **200** at an elevated temperature or room temperature, and a first fire operation using the voltage pulse V.sub.FF with a second width W1 of 40 ns is performed on another test structure **200** at the same temperature. After performing these first fire operations, off currents of the selectors **110** in the test structures **200** subjected to these first fire operations are measured. A fixed voltage low enough to prevent turning on the selectors **110** is provided to these test structures **200**, and the off currents of the selectors **110** can be observed by reading the currents IR. For each of these test structures **200**, the off current is measured multiple times, and the results are shown in FIG. 2D.

[0054] Referring to FIG. 2D, 25.sup.th percentile to 75.sup.th percentile of the off currents measured from the test structure **200** subjected to the first fire operation using the voltage pulse V.sub.FF having the width W1 of 20 μ s are represented by an error bar, and a median value of these

off currents is represented by a data point. Similarly, 25.sup.th percentile to 75.sup.th percentile of the off currents measured from the test structure **200** subjected to the first fire operation using the voltage pulse $V_{sub,FF}$ having the width $W1$ of 40 ns are represented by an error bar, and a median value of these off currents is represented by a data point. As indicated by the median values, the off current of the selector **110** decreases as the voltage pulse $V_{sub,FF}$ used in the first fire operation has a shorter width $W1$. Further, as indicated by the error bars, variation of the off current of the selector **110** is reduced when the voltage pulse $V_{sub,FF}$ used in the first fire operation has a shorter width $W1$, and this result also implies that variation of the threshold voltage $V_{sub,TH}$ of the selector **110** can be reduced when the width $W1$ of the voltage pulse $V_{sub,FF}$ used in the first fire operation is shortened.

[0055] It should be noted that, the first fire operation according to various embodiments is described by using a triangular pulse (e.g., the voltage pulse $V_{sub,FF}$ as shown in FIG. 2A). However, a voltage pulse with other pulse shape (e.g., a rectangular pulse) is also available, as long as a pulse width is within the described range.

[0056] FIG. 3A is a flow diagram illustrating an adaptive first fire operation **300** performed on a memory cell **100**, according to some embodiments of the present disclosure. FIG. 3B illustrates input voltage pulses $P2$ and resulted output current during the adaptive first fire operation as shown in FIG. 3A.

[0057] Referring to FIG. 3A and FIG. 3B, the adaptive first fire operation **300** is provided for further reducing resulted electrical stress. The adaptive first fire operation **300** is initially performed on one of the memory cell **100** at an elevated temperature (e.g., ranging from about 45° C. to about 150° C.) or room temperature, and is repeated on next memory cell **100** until all of the memory cells **100** are initialized. In each time, the adaptive first fire operation **300** may include successively providing multiple voltage pulses $P2$ to the selector **110**. A peak amplitude $VP2$ of each voltage pulse $P2$ is lower than a peak amplitude $VP2$ of the next voltage pulse $P2$. As an example, each voltage pulse $P2$ may be a square pulse. However, each voltage pulse $P2$ may alternatively be a triangular pulse similar to the voltage pulse $P1$ described with reference to FIG. 2A, or a voltage pulse with any pulse shape.

[0058] The adaptive first fire operation **300** begins with an operation **S300**, and ends at an operation **S308**. A cycle including operations between the opening operation **S300** and the ending operation **S308** is performed at least once. At the operation **S300**, a peak amplitude $VP2$ of a first one of the voltage pulse $P2$ is set to be a pre-determined minimum peak amplitude $VP2_{sub,min}$. Thereafter, at an operation **S302**, the voltage pulse $P2$ is provided to the selector **110**, then an output current passing through the selector **110** is sensed during an operation **S304**. For instance, a sense amplifier may be used for sensing the output current.

[0059] At an operation **S306**, the peak amplitude $VP2$ is compared to a pre-determined maximum peak amplitude $VP2_{sub,max}$. The maximum peak amplitude $VP2_{sub,max}$ is high enough that formation of the initial conductive path in the selector **110** can be ensured, but might be greater than the actual first fire voltage $V_{sub,FF}$ of the selector **110**. If the peak amplitude $VP2$ reaches the maximum peak amplitude $VP2_{sub,max}$, then the initialization of the selector **110** is considered to be completed, and is ended as indicated by an operation **S308**.

[0060] On the other hand, if the peak amplitude $VP2$ does not reach the maximum peak amplitude $VP2_{sub,max}$, whether if the selector **110** has been initialized needs a further check. Accordingly, as indicated by an operation **S310**, another comparison is performed. During the operation **S310**, the output current is compared to a pre-determined target current $I_{sub,target}$. The target current $I_{sub,target}$ is set to be higher than the ground level of the output current (i.e., an off-current of the selector **110**), and may be equal to or slightly greater than a current at the threshold voltage $V_{sub,TH}$ as shown in FIG. 1D, but lower than an on-current of the selector **110** (the current at the stage **S3** as shown in FIG. 1D). That is, if the output current reaches the target current $I_{sub,target}$, then the initialization of the selector **110** is considered to be completed, and should be ended as

indicated by an operation **S308**.

[0061] If the output current does not reach the target current $I_{sub.target}$, it indicates that the selector **110** has not been initialized. In other words, the peak amplitude $VP2$ of the voltage pulse **P2** to be provided in next cycle has to be increased, as indicated by an operation **S312**. In some embodiments, the peak amplitude $VP2$ of the next voltage pulse **P2** is set to be increased by an increment $VP2.sub.inc$. The cycle including the operations between the opening operation **S300** and the ending operation **S308** is repeated until the peak amplitude $VP2$ reaches the maximum peak amplitude $VP2.sub.max$ or the output current reaches the target current $I_{sub.target}$. As an example as shown in FIG. 3B, the output current reaches the target current $I_{sub.target}$ in the third cycle, in which the peak amplitude $VP2$ is still lower than the maximum peak amplitude $VP2.sub.max$. Accordingly, the first fire operation **300** is terminated at the end of the third cycle, thus the voltage pulses and the output current in the following cycles are depicted by dash lines.

[0062] By properly setting the minimum peak amplitude $VP2.sub.min$ and the increment $VP2.sub.inc$, the peak amplitude $VP2$ of the voltage pulse **P2** used in the final cycle (i.e., the cycle that initialize the selector **110**) can be as close to the actual first fire voltage $V_{sub.FF}$ as possible. Therefore, the adaptive fire operation **300** can effectively limit electrical stress resulted on the selector **110**.

[0063] FIG. 4A is a flow diagram illustrating an adaptive first fire operation **400** performed on a memory cell **100**, according to some embodiments of the present disclosure. FIG. 4B illustrates input voltage pulses **P2**, **P3** and resulted output current during the adaptive first fire operation as shown in FIG. 4A.

[0064] Referring FIG. 4A and FIG. 4B, the first fire operation **400** is adaptive based on off current variation of the selector **110**. As similar to the adaptive first fire operation **300** described with reference to FIG. 3A and FIG. 3B, the adaptive first fire operation **400** is initially performed on one of the memory cell **100** at an elevated temperature (e.g., ranging from about 45° C. to about 150° C.) or room temperature, and is repeated on next memory cell **100** until all of the memory cells **100** are initialized.

[0065] For each memory cell **100**, groups of the voltage pulses **P2** are successively provided. The voltage pulses **P2** in each group have the same peak amplitude $VP2$, and the peak amplitude $VP2$ of the voltage pulses **P2** in each group is lower than the peak amplitude $VP2$ of the voltage pulses **P2** in a next group. Further, each voltage pulse **P2** is followed by a voltage pulse **P3** for reading the off current of the selector **110**. The voltage pulses **P3** may have identical peak amplitude $VP3$, which is much lower than the peak amplitude $VP2$ of the voltage pulses **P2** (e.g., much lower than the minimum peak amplitude $VP2.sub.min$ of the voltage pulses **P2** in the first group), for monitoring the off current of the selector **110** when the selector **110** is in an off state (i.e., at the stage **S1** as described with reference to FIG. 1D). It should be noted that, a bottommost diagram in FIG. 4B is a zoomed-in view of the off current shown in a middle diagram in FIG. 4B. In the bottommost diagram in FIG. 4B, portions of the output current in response to the voltage pulses **P2** are omitted.

[0066] The adaptive first fire operation **400** begins with an operation **S400**, and ends at an operation **S408**. A cycle including operations between the opening operation **S400** and the ending operation **S408** is performed at least once. At the opening operation **S400**, a peak amplitude $VP2$ of a voltage pulse **P2** is set to be the minimum peak amplitude $VP2.sub.min$, and a repetition number of each of the voltage pulse **P2** and a following voltage pulse **P3** is set as 1. Thereafter, at an operation **S402**, the voltage pulse **P2** and the following voltage pulse **P3** are provided to the selector **110**, and then an output current passing through the selector **110** is sensed during an operation **S404**.

[0067] At a subsequent operation **S406**, whether if the peak amplitude $VP2$ of the voltage pulse **P2** reaches the maximum peak amplitude $VP2.sub.max$ is determined, and whether if a repetition number of the voltage pulses **P2**, **P3** reaches a pre-determined target value (e.g., 3 times) is determined. If the peak amplitude $VP2$ reaches the maximum peak amplitude $VP2.sub.max$ and the

repetition number reaches the target value, then the initialization of the selector **110** is considered to be completed, and should be ended as indicated by an operation **S408**. On the other hand, if the peak amplitude VP2 does not reach the maximum peak amplitude VP2.sub.max or the repetition number does not reach the target value, another comparison is performed at an operation **S410**. [0068] During the operation **S410**, whether if the off current in response to the voltage pulse P3 reaches a target off current I.sub.off, target is determined. The target off current I.sub.off, target may be set to be a value equal to or slightly higher than a critical off current that is resulted from the voltage pulse P2 with the peak amplitude VP2 just high enough to initialize the selector **110**. That is, the target off current I.sub.off, target may be indicative of initialization of the selector **110**. If the off current reaches the target off current I.sub.off, target, then the first fire operation is considered to be completed, and should be ended as indicated by the operation **S408**. On the other hand, if the off current is still below the target off current I.sub.off, target, the first fire operation **400** goes to the operation **S412**.

[0069] At the operation **S412**, whether if the repetition number reaches the target value is determined, for deciding whether to increase the peak amplitude VP2 of the voltage pulse P2 in the next cycle. If the repetition number reaches the target value, then the peak amplitude VP2 of the voltage pulse P2 to be provided in next cycle is required to be increased, as indicated by an operation **S414**. In some embodiments, the peak amplitude VP2 of the next voltage pulse P2 is set to be increased by an increment VP2.sub.inc. On the other hand, if the repetition number does not reach the target value, then the voltage pulse P2 (with the same peak amplitude VP2) and the following voltage pulse P3 are repeated once again in the next cycle. That is, the repetition number is increased by 1, as indicated by an operation **S416**. The next cycle may begin with the operation **S402**, for providing the voltage pulse P2 and the following voltage pulse P3 to the selector **110**.

[0070] The cycle including the operations between the opening operation **S400** and the ending operation **S408** is repeated until the peak amplitude VP2 reaches the maximum peak amplitude VP2.sub.max or the off current reaches the target off current I.sub.off, target.

[0071] As an example as shown in FIG. 4B, the off current reaches the target off current I.sub.off, target in the fifth cycle, in which the peak amplitude VP2 is still lower than the maximum peak amplitude VP2.sub.max. Accordingly, the first fire operation **400** is terminated at the end of the fifth cycle, thus the voltage pulses and the output current in the following cycles are depicted by dash lines. Further, as shown in FIG. 4B, the output current in response to the voltage pulses P2, P3 significantly increases since the fourth cycle, which indicates that the selector **110** may be initialized during the fourth cycle.

[0072] By providing the voltage pulses P3 to the selector **110**, the off current of the selector **110** can be monitored. Further, by properly adjusting the target off current I.sub.off, target, the selector **110** can be initialized without being subjected to excessive electrical stress.

[0073] FIG. 5A is a schematic diagram illustrating an adaptive first fire operation **500**, according to some embodiments of the present disclosure. FIG. 5B is a flow diagram illustrating the adaptive first fire operation **500**, according to some embodiments of the present disclosure. FIG. 5C illustrates input voltage pulses P2, P4 and resulted output current during the adaptive first fire operation **500** as shown in FIG. 5B.

[0074] Referring to FIG. 5A, during the adaptive first fire operation **500**, each memory cell **100** is coupled to a field effect transistor **502**. As similar to the adaptive first fire operation **300** described with reference to FIG. 3A and FIG. 3B, the adaptive first fire operation **500** includes providing voltage pulses P2 to the selector **110**, and sensing output current passing through the selector **110**. In addition, the adaptive first fire operation **500** further includes controlling the field effect transistor **502** to limit the output current. A source/drain terminal of the field effect transistor **502** is coupled to a terminal of the resistance variable storage element **120** that is not shared with the selector **110**, and the other source/drain terminal of the field effect transistor **502** is coupled to a voltage source (e.g., ground voltage) through a, for example, sense amplifier (not shown). In

addition, a gate terminal of the field effect transistor **502** is configured to receive the voltage pulses **P4**. In response to the voltage pulses **P4**, the field effect transistor **502** is turned on and operated in a saturation region as a result of a voltage difference between the source/drain terminals. In this way, the output current is limited to a saturation current of the field effect transistor **502**. By controlling a peak amplitude **VP4** of the voltage pulses **P4**, the saturation current and thereby the output current can be adjusted. By limiting the output current, the electrical stress resulted on the selector **110** can be further reduced. In some embodiments, each memory cell **100** is coupled to a respective field effect transistor **502**. In other embodiments, multiple memory cells **100** (e.g., a row/column of memory cells **100**) are coupled to a same field effect transistor **502**.

[0075] The adaptive first fire operation **500** is initially performed on one of the memory cell **100** at an elevated temperature (e.g., ranging from about 45° C. to about 150° C.) or room temperature, and is repeated on next memory cell **100** until all of the memory cells **100** are initialized. In each time, the adaptive first fire operation **500** may include successively providing a plurality of the voltage pulses **P2** to the selector **110**. Meanwhile, a plurality of the voltage pulses **P4** are provided to the field effect transistor **502**.

[0076] Referring to FIG. 5A and FIG. 5B, the adaptive first fire operation **500** begins with operations **S500a**, **S500b**, and ends at an operation **S508**. A cycle including operations between the opening operations **S500a**, **S500b** and the ending operation **S508** is performed at least once. At the operation **S500a**, the peak amplitude **VP2** of the first voltage pulse **P2** is set to be the pre-determined minimum peak amplitude **VP2.sub.min**. Meanwhile, at the operation **S500b**, a peak amplitude of the voltage pulses **P4** is set to the peak amplitude **VP4**, which is able to turn on the field effect transistor **502**. Thereafter, at an operation **S502**, the voltage pulse **P2** is provided to the selector **110**, and one of the voltage pulses **P4** is simultaneously provided to the gate terminal of the field effect transistor **502** for turning on the field effect transistor **502**. Subsequently, an output current passing through the selector **110** and the field effect transistor **502** is sensed during an operation **S504**. For instance, a sense amplifier coupled to one of the source/drain terminals of the field effect transistor **502** may be used for sensing the output current.

[0077] At an operation **S506**, the peak amplitude **VP2** is compared to the pre-determined maximum peak amplitude **VP2.sub.max**. The maximum peak amplitude **VP2.sub.max** is set as high enough that formation of the initial conductive path in the selector **110** can be ensured, but might be greater than the actual first fire voltage **V.sub.FF** of the selector **110**. If the peak amplitude **VP2** reaches the maximum peak amplitude **VP2.sub.max**, then the initialization of the selector **110** is considered to be completed, and is ended as indicated by an operation **S508**.

[0078] On the other hand, if the peak amplitude **VP2** does not reach the maximum peak amplitude **VP2.sub.max**, whether if the selector **110** has been initialized needs a further check. Accordingly, as indicated by an operation **S510**, another comparison is performed. During the operation **S510**, the output current is compared to the pre-determined target current **I.sub.target**. The target current **I.sub.target** is set to be higher than the ground level of the output current, and may be equal to or slightly greater than a current at the threshold voltage **V.sub.TH** as shown in FIG. 1D. That is, if the output current reaches the target current **I.sub.target**, then the initialization of the selector **110** is considered to be completed, and should be ended as indicated by an operation **S508**.

[0079] If the output current does not reach the target current **I.sub.target**, it indicates that the selector **110** has not been initialized. In other words, the peak amplitude **VP2** of the voltage pulse **P2** to be provided in next cycle has to be increased, as indicated by an operation **S512**. In some embodiments, the peak amplitude **VP2** of the next voltage pulse **P2** is set to be increased by the increment **VP2.sub.inc**. The cycle including the operations between the opening operation **S500a**, **S500b** and the ending operation **S508** is repeated until the peak amplitude **VP2** reaches the maximum peak amplitude **VP2.sub.max** or the output current reaches the target current **I.sub.target**.

[0080] As an example as shown in FIG. 5B, the output current reaches the target current **I.sub.target** in the third cycle, in which the peak amplitude **VP2** is still lower than the maximum

peak amplitude $VP2.sub.max$. Accordingly, the first fire operation **300** is terminated at the end of the third cycle, thus the voltage pulses and the output current in the following cycles are depicted by dash lines. In addition, the voltage pulses **P4** are provided to the field effect transistor **502** at the same time that the voltage pulses **P2** are provided to the selector **110**, such that the field effect transistor **502** is turned on when the selector **110** receives the voltage pulses **P2**. Accordingly, even when the output current abruptly increases as the selector **110** is switched on, the output current can be limited to a saturation current $I.sub.sat$ of the field effect transistor **502**, which may be greater than the target current $I.sub.target$.

[0081] By properly setting the minimum peak amplitude $VP2.sub.min$ and the increment $VP2.sub.inc$, the peak amplitude $VP2$ of the voltage pulse **P2** used in the final cycle (i.e., the cycle that initialize the selector **110**) can be as close to the actual first fire voltage $V.sub.FF$ as possible. On top of that, the on current passing through the selector **110** can be limited by the field effect transistor **502**. Therefore, the adaptive fire operation **500** can effectively limit electrical stress resulted on the selector **110**.

[0082] Although not shown, the adaptive first fire operation **400** as described with reference to FIG. 4A and FIG. 4B may further include coupling the memory cell **100** to a field effect transistor for limiting the output current, as similar to the adaptive first fire operation **500** as described with reference to FIG. 5A through FIG. 5C.

[0083] FIG. 6 is a schematic cross-sectional view illustrating a semiconductor chip **600** with the memory array **10** (shown in FIG. 1A) embedded therein, according to some embodiments of the present disclosure.

[0084] Referring to FIG. 6, the semiconductor chip **600** may have a front-end-of-line (FEOL) structure **600F** built on a semiconductor substrate **602**. The FEOL structure **600F** may include transistors **604**. Each of the transistors **604** may include a gate structure **606** and a pair of source/drain structures **608** at opposite sides of the gate structure **606**. Further, adjacent transistors **604** may be isolated from one another by an isolation structure **610** formed in the semiconductor substrate **602**. In some embodiments, the transistors **604** are planar type transistors. In these embodiments, the gate structures **606** are formed on planar portions of the semiconductor substrate **602**, and the source/drain structures **608** may be formed in the semiconductor substrate **602**. In alternative embodiments, the transistors **604** are fin type transistors or gate-all-around (GAA) transistors. In these alternative embodiments, the semiconductor substrate **602** may be shaped to form fin structures at its top surface, or stacks of channel structures (e.g., stacks of semiconductor nanosheets) may be formed on the semiconductor substrate **602**. Each fin structure/channel structure may extend between a pair of the source/drain structures **608**. In addition, the gate structures **606** may intersect and cover the fin structures or the stacks of channel structures. Moreover, the FEOL structure **600F** may further include a dielectric layer **612** and contact structures **614** formed in the dielectric layer **612**. The contact structures **614** penetrate through the dielectric layer **612**, to establish electrical contact with the source/drain structures **608**.

[0085] The memory array **10** may be embedded in a back-end-of-line (BEOL) structure **600B** formed on the FEOL structure **600F**. The BEOL structure **600B** may include a stack of interlayer dielectric layers **616**. For conciseness, only one of the interlayer dielectric layers **616** is labeled. The memory array **10** is formed in successive ones of the interlayer dielectric layers **616**. Further, the BEOL structure **600B** also includes conductive elements **618** spreading in the stack of interlayer dielectric layers **616**, for interconnecting the transistors **604**, and for out-routing the memory array **10**. The conductive elements **618** may be distributed below, around and over the memory array **10**, and may include conductive patterns **620** and conductive vias **622**. Each conductive pattern **620** laterally extends in one of the interlayer dielectric layers **616**. In addition, each conductive via **622** vertically extends through one or more of the interlayer dielectric layers **616** to establish electrical contact with one or more of the conductive patterns **620**, or to establish electrical contact with one of the signal lines in the memory array **10** (i.e., one of the first signal

lines SL1 or one of the second signal lines SL2). The memory array **10** may be routed to some of the transistors **604** in the FEOL structure **600F** through some of the conductive elements **618**, and can be driven by a driving circuit including these transistors **604**.

[0086] Although the conductive elements **618** below the memory array **10** are depicted as being connected to the conductive elements **618** above the memory array **10** through a long conductive via **622** penetrating through multiple interlayer dielectric layers **616**, such long conductive via **622** may be alternatively replaced by a combination of short conductive vias **622** and some conductive patterns **620**. The present disclosure is not limited to the arrangement of the conductive elements **618**. Further, although not shown, passivation layer(s) and electrical connectors as chip inputs/outputs (I/Os) may be formed on the BEOL structure **600B**.

[0087] FIG. 7 is a schematic three-dimensional view illustrating a memory array **10a**, according to some embodiments of the present disclosure.

[0088] The memory array **10** as shown in FIG. 1A includes the memory cells **100** arranged at a single horizontal level. On the other hand, the memory array **10a** as shown in FIG. 7 has multiple horizontal levels. As shown in FIG. 7, layers of the memory cells **100** are stacked along a vertical direction. In addition, each layer of the memory cells **100** are coupled to vertically separated layers of signal lines (i.e., a layer of first signal lines SL1 and a layer of second signal lines SL2). Layers of the first signal lines SL1 and layers of the second signal lines SL2 may be alternatively arranged along a vertical direction. Accordingly, the memory cells **100** in some layers are coupled to underlying first signal lines SL1 and overlying second signal lines SL2, while the memory cells **100** in other layers are coupled to underlying second signal lines SL2 and overlying first signal lines SL1. In addition, vertically adjacent layers of the memory cells **100** may share the layer of signal lines extending in between. For instance, the bottommost layer of the memory cells **100** and the second bottommost layer of the memory cells **100** may share a layer of the second signal lines SL2 in between.

[0089] As being deployed along the vertical direction, the memory array **10a** is no longer limited to two-dimensional design, and storage density can be significantly increased without increasing a footprint area of the memory array **10a**. Each horizontal level of the memory array **10a** may be defined by a layer of the memory cells **100** and the layers of the first and second signal lines SL1, SL2 connected thereto. Although the memory array **10a** is depicted as having four horizontal levels, those skilled in the art may adjust an amount of the horizontal levels of the memory array **10a**. For instance, the memory cell **10a** may have two to ten horizontal levels. Further, as similar to the memory array **10** described with reference to FIG. 6, the memory array **10a** shown in FIG. 7 may be embedded in a BEOL structure of a semiconductor chip as well.

[0090] FIG. 8A is a cross-sectional view schematically illustrating a memory cell **100a**, according to some embodiments of the present disclosure. The memory cell **100a** is similar to the memory cell **100** as described with reference to FIG. 1B. Therefore, only differences between the memory cells **100**, **100a** will be described. The same or the like parts between the memory cells **100**, **100a** would not be repeated again.

[0091] Referring to FIG. 8A, a resistance variable storage element **120a** in a memory cell **100a** may use one of the signal lines SL1, SL2 coupled to the memory cell **100a** as a bottom terminal, while the electrode **114** may be functioned as a top terminal of the resistance variable storage element **120a**, which is shared with the overlying selector **110**. In addition, a storage layer **124a** lying below the electrode **114** may further extend to the signal line as the bottom terminal of the resistance variable element **120a**. For instance, as shown in FIG. 8A, the first signal line SL1 is functioned as the bottom terminal of the resistance variable storage element **120a**, while the electrode **114** is functioned as the top terminal of the resistance variable storage element **120a**. In addition, the storage layer **124a** may have a horizontal portion lying below the electrode **114**, and have a vertical portion extending from the horizontal portion to the first signal line SL1. Although not shown, the resistance variable storage element **120a** in another memory cell **100a** may use a

second signal line SL2 as a bottom terminal, and the electrode **114** may be functioned as a top terminal of this memory cell **100a**. In addition, the storage layer **124a** may extend to the second signal line SL2 along a vertical direction.

[0092] Further, as similar to the memory cell **100** as described with reference to FIG. 1A and FIG. 7, a plurality of the memory cells **100a** may be arranged as a two-dimensional memory array or a three-dimensional memory array. Moreover, the two-dimensional memory array or the three-dimensional memory array may be embedded in a BEOL structure of a semiconductor chip, as described with reference to FIG. 6.

[0093] FIG. 8B is a cross-sectional view schematically illustrating a memory cell **100b**, according to some embodiments of the present disclosure. The memory cell **100b** is similar to the memory cell **100** as described with reference to FIG. 1B. Therefore, only differences between the memory cells **100**, **100b** will be described. The same or the like parts between the memory cells **100**, **100b** would not be repeated again.

[0094] Referring to FIG. 8B, a resistance variable storage element **120b** is stacked on a selector **110b** in a memory cell **100b**. Electrodes **800**, **802** at opposite sides of the switching layer **116** are functioned as bottom and top terminals of the selector **110b**, respectively. The electrode **800** lying below the switching layer **116** and functioned as the bottom terminal of the selector **110b** may extend to the signal line extending below the memory cell **100b** (e.g., a first signal line SL1). On the other hand, the electrode **802** disposed on the switching layer **116** and functioned as the top terminal of the selector **110b** may be shared with the resistance variable storage element **120b**. In addition to the electrode **802** lying below the storage layer **124** and functioned as a bottom terminal of the resistance variable storage element **120**, the resistance variable storage element **120b** may include an electrode **804** located on the storage layer **124** and functioned as its top terminal. The electrode **804** may extend from a top side of the storage layer **124** to the signal line extending on the memory cell **100b** (e.g., a second signal line SL2). In some embodiments, the memory cell **100b** is formed in a pillar shape. In these embodiments, sidewalls of the electrodes **800**, **802**, **804**, the switching layer **116** and the storage layer **124** may be substantially coplanar with one another. In addition, a sidewall of such pillar structure is substantially flat from a top end of the pillar structure to a bottom end of the pillar structure. Further, the electrodes **800**, **802**, **804** may be similar to the electrodes **112**, **114**, **122** in terms of material candidates.

[0095] Further, as similar to the memory cell **100** as described with reference to FIG. 1A and FIG. 7, a plurality of the memory cells **100b** may be arranged as a two-dimensional memory array or a three-dimensional memory array. Moreover, the two-dimensional memory array or the three-dimensional memory array may be embedded in a BEOL structure of a semiconductor chip, as described with reference to FIG. 6.

[0096] As above, improved first fire operations used for an OTS selector are provided according to various embodiments. In some embodiments, a first fire operation using an AC pulse is performed at an elevated temperature. In these embodiments, additional energy may be provided for overcoming the activation barrier for forming the initial conductive path in the OTS selector, such that the peak amplitude of the voltage pulse used in the first fire operation (which is indicative of a first fire voltage of the OTS selector) can be lowered. As a result, the OTS selector used in an embedded memory can be more compatible with logic circuits. On top of that, since the voltage pulse is provided as an AC pulse with extremely short width, electrical stress resulted on the OTS selector can be further reduced, such that off current variation of the OTS selector can be effectively limited. In other embodiments, the first fire operation is adaptive, that a series of voltage pulse with incremental peak amplitudes are successively provided to the OTS selector, and the first fire operation is determined to be ended when the input voltage provided to the OTS selector just sufficient for overcoming the activation barrier. In these embodiments, electrical stress resulted on the OTS selector and thus the off current variation of the OTS selector resulted from the first fire operation can be more precisely controlled.

[0097] In an aspect of the present disclosure, a first fire operation for an OTS selector is provided. The first fire operation comprises: setting a peak amplitude of a voltage pulse; and performing at least one cycle, comprising: providing the voltage pulse to the OTS selector; sensing an output current passing through the OTS selector in response to the received voltage pulse; comparing a peak amplitude of the voltage pulse with a maximum peak amplitude ensuring initialization of the OTS selector; ending the first fire operation if the peak amplitude reaches the maximum peak amplitude; comparing the output current with a target current indicative of initialization of the OTS selector if the peak amplitude is lower than the maximum peak amplitude; ending the first fire operation if the output current reaches the target current; and setting another voltage pulse with a greater peak amplitude if the output current is lower than the target current.

[0098] In another aspect of the present disclosure, a first fire operation for an OTS selector is provided. The first fire operation comprises: setting a repetition number of a first fire voltage pulse and a read voltage pulse as one; and performing at least two cycles, each comprising: sequentially providing the first fire voltage pulse and the read voltage pulse to the OTS selector; sensing an output current passing through the OTS selector in response to the first fire voltage pulse and the read voltage pulse; comparing a peak amplitude of the voltage pulse with a maximum peak amplitude ensuring initialization of the OTS selector, and comparing the repetition number of the first fire voltage pulse and the read voltage pulse with a target repetition number; ending the first fire operation if the peak amplitude reaches the maximum peak amplitude and the repetition number reaches the target repetition number; comparing the output current in response to the read voltage pulse with a target off current if the peak amplitude is lower than the maximum peak amplitude or the repetition number is lower than the target repetition number; ending the first fire operation if the output current in response to the read voltage pulse reaches the target off current; comparing the repetition number with the target repetition number if the output current in response to the read voltage pulse is lower than the target off current; setting a next first fire voltage pulse with a greater peak amplitude if the repetition number reaches the target repetition number; and increasing a repetition number of the next first fire voltage pulse and a following read voltage pulse by one.

[0099] In yet another aspect of the present disclosure, a first fire operation for an OTS selector is provided. The OTS selector is coupled to a transistor. The first fire operation comprises: setting a peak amplitude of a first fire voltage pulse and a peak amplitude of a gate voltage pulse; and performing at least one cycle, comprising: providing the first fire voltage pulse to the OTS selector and providing the gate voltage pulse to the transistor for turning on the transistor; sensing an output current passing through the OTS selector in response to the first fire voltage pulse; comparing a peak amplitude of the first fire voltage pulse with a maximum peak amplitude ensuring initialization of the OTS selector; ending the first fire operation if the peak amplitude reaches the maximum peak amplitude; comparing the output current with a target current indicative of initialization of the OTS selector if the peak amplitude is lower than the maximum peak amplitude; ending the first fire operation if the output current reaches the target current; and setting another first fire voltage pulse with a greater peak amplitude if the output current is lower than the target current.

[0100] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A first fire operation for an ovonic threshold switch (OTS) selector, comprising: during the first fire operation, applying a plurality of voltage pulses with gradually increasing peak amplitudes in sequence to the OTS selector; stopping the first fire operation when a peak amplitude of a first voltage pulse reaches a pre-determined maximum value, or when an output current of the OTS selector in response to a second voltage pulse reaches a target value.
2. The first fire operation for the OTS selector according to claim 1, wherein a difference from the peak amplitude of each voltage pulse to the peak amplitude of a next one of the voltage pulses is constant throughout the series of the voltage pulses.
3. The first fire operation for the OTS selector according to claim 1, wherein determining whether if the output current reaches the target value is performed every time when the OTS selector receives one of the voltage pulses.
4. The first fire operation for the OTS selector according to claim 1, wherein the OTS selector receives one or more of the voltage pulses before end of the first fire operation.
5. The first fire operation for the OTS selector according to claim 1, wherein each voltage pulse is a rectangular pulse or a triangular pulse.
6. The first fire operation for the OTS selector according to claim 1, wherein the voltage pulses are identical with one another except for peak amplitude.
7. The first fire operation for the OTS selector according to claim 1, wherein a pulse width of each voltage pulse ranges from 40 nanoseconds to 20 microseconds.
8. The first fire operation for the OTS selector according to claim 1, wherein the first fire operation is performed at an elevated temperature ranging from about 45° C. to about 150° C.
9. A first fire operation for an OTS selector, comprising: applying a series of first voltage pulses and a plurality of second voltage pulses respectively following one of the first voltage pulses to the OTS selector, until the peak amplitude of a latest one of the first voltage pulses reaches a pre-determined maximum peak amplitude, or until an off current of the OTS selector in response to a latest one of the second voltage pulses reaches a target off current, wherein peak amplitudes of two adjacent first voltage pulse and second voltage pulse are different.
10. The first fire operation for the OTS selector according to claim 9, wherein the first voltage pulses are provided by groups, the first voltage pulses in each group are identical with one another, and the peak amplitude of the first voltage pulses in each of the groups is greater than the peak amplitude of the first voltage pulses in a previous one of the groups.
11. The first fire operation for the OTS selector according to claim 9, wherein output currents of the OTS selector in response to the first voltage pulses are not used for determining end of the first fire operation.
12. The first fire operation for the OTS selector according to claim 9, wherein the pre-determined maximum peak amplitude is set to be sufficiently high to ensure initialization of the OTS selector.
13. The first operation for the OTS selector according to claim 9, wherein the peak amplitude of the second voltage pulses is set to be lower than a threshold voltage of the OTS selector.
14. The first fire operation for the OTS selector according to claim 9, wherein the first fire operation is performed at an elevated temperature ranging from about 45° C. to about 150° C.
15. The first fire operation for the OTS selector according to claim 9, wherein a pulse width of each first voltage pulse ranges from 40 nanoseconds to 20 microseconds.
16. A first fire operation for an OTS selector, comprising: providing a series of first voltage pulses, wherein a peak amplitude of each first voltage pulse is higher than a peak amplitude of a previous one of the first voltage pulses; and sequentially sending the first voltage pulses to the OTS selector through a field effect transistor, until the peak amplitude of a latest one of the first voltage pulses reaches a pre-determined maximum value, or until an output current passing through the OTS

selector in response to a latest one of the first voltage pulses reaches a target value, wherein the field effect transistor and the OTS selector are coupled in series.

17. The first fire operation for the OTS selector according to claim 16, wherein the field effect transistor is coupled to the OTS selector by a source/drain terminal.

18. The first fire operation for the OTS selector according to claim 16, further comprising: sending second voltage pulses to a gate terminal of the field effect transistor at the same time the first voltage pulses are sent to the OTS selector.

19. The first fire operation for the OTS selector according to claim 16, wherein the first fire operation is performed at an elevated temperature ranging from about 45° C. to about 150° C.

20. The first fire operation for the OTS selector according to claim 16, wherein a pulse width of each first voltage pulse ranges from 40 nanoseconds to 20 microseconds.
